Unit 1: Mechanics and Materials

IAS compulsory unit

Externally assessed

1.1 Unit description

Introduction	This unit covers mechanics and materials. This topic may be studied using applications that relate to mechanics, for example sports and to materials, for example spare-part surgery. This topic also enables students to develop practical and mathematical skills.
Practical skills	In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include strobe photography or the use of a video camera to analyse projectile motion, determine the centre of gravity of an irregular rod, investigate the conservation of momentum using light gates and air track, Hooke's law and the Young modulus experiments for a variety of materials.
Mathematical skills	Mathematical skills that could be developed in this topic include: plotting two variables from experimental data; calculating rate of change from a graph showing a linear relationship; drawing and using the slope of a tangent to a curve as a measure of rate of change; calculating or estimating, by graphical methods as appropriate, the area between a curve and the x-axis and realising the physical significance of the area that has been determined; distinguishing between instantaneous rate of change and average rate of change and identifying uncertainties in measurements; using simple techniques to determine uncertainty when data are combined; using angles in regular 2D and 3D structures with force diagrams and using sin, cos and tan in physical problems.